

Sequence diagram

Sequence Diagram

- In [software engineering](#), a **sequence diagram** shows process interactions arranged in time sequence. This diagram depicts the processes and objects involved and the sequence of messages exchanged as needed to carry out the functionality.

A system sequence diagram should specify and show the following:

- External actors
- Messages (methods) invoked by these actors
- Return values (if any) associated with previous messages
- Indication of any loops or iteration area

Use of Sequence Diagrams

Sequence diagrams are used because they clearly show how different objects or components in a system interact with each other in a specific order over time. They focus on the sequence of messages exchanged between system elements.

1. Understanding Dynamic Behavior

Sequence diagrams show how objects interact step-by-step, helping us understand how a system works during execution.

2. Better Communication

They provide a simple and visual way to explain system behavior, making it easier for team members to understand complex processes without looking at the actual code.

3. Use Case Representation

Sequence diagrams help represent use cases by showing how a particular function or process is carried out in the system.

Use of Sequence Diagrams

4. System Design Support

They help in designing system architecture by showing how different components or services communicate, especially in large or distributed systems.

5. Documentation

Sequence diagrams act as proper documentation, showing how different parts of the system interact. This helps developers and maintenance teams in the future.

6. Error Detection and Debugging

By showing the flow of interactions clearly, sequence diagrams help identify delays, mistakes, or inefficiencies in the system.

Sequence diagram - Elements

Sequence Diagrams show elements as they interact over time and they are organized according to object (horizontally) and time (vertically):

Object Dimension: The horizontal axis shows the elements that are involved in the interaction.

Time Dimension: The vertical axis represents time proceedings (or progressing) down the page.

Elements of Sequence Diagram

- Label

The UML specification provides specific text values for diagram types (e.g., sd = Sequence Diagram, activity = Activity Diagram, and use case = Use Case Diagram).

Lifeline

When drawing a sequence diagram, lifeline notation elements are placed across the top of the diagram. Lifelines represent either roles or object instances that participate in the sequence being modelled.

- Messages

To show an object (i.e., lifeline) sending a message to another object, draw a line to the receiving object with a solid arrowhead (if a synchronous call operation) or with a stick arrowhead (if an asynchronous signal). The message/method name is placed above the arrowed line. The message that is being sent to the receiving object represents an operation/method that the receiving object's class implements.

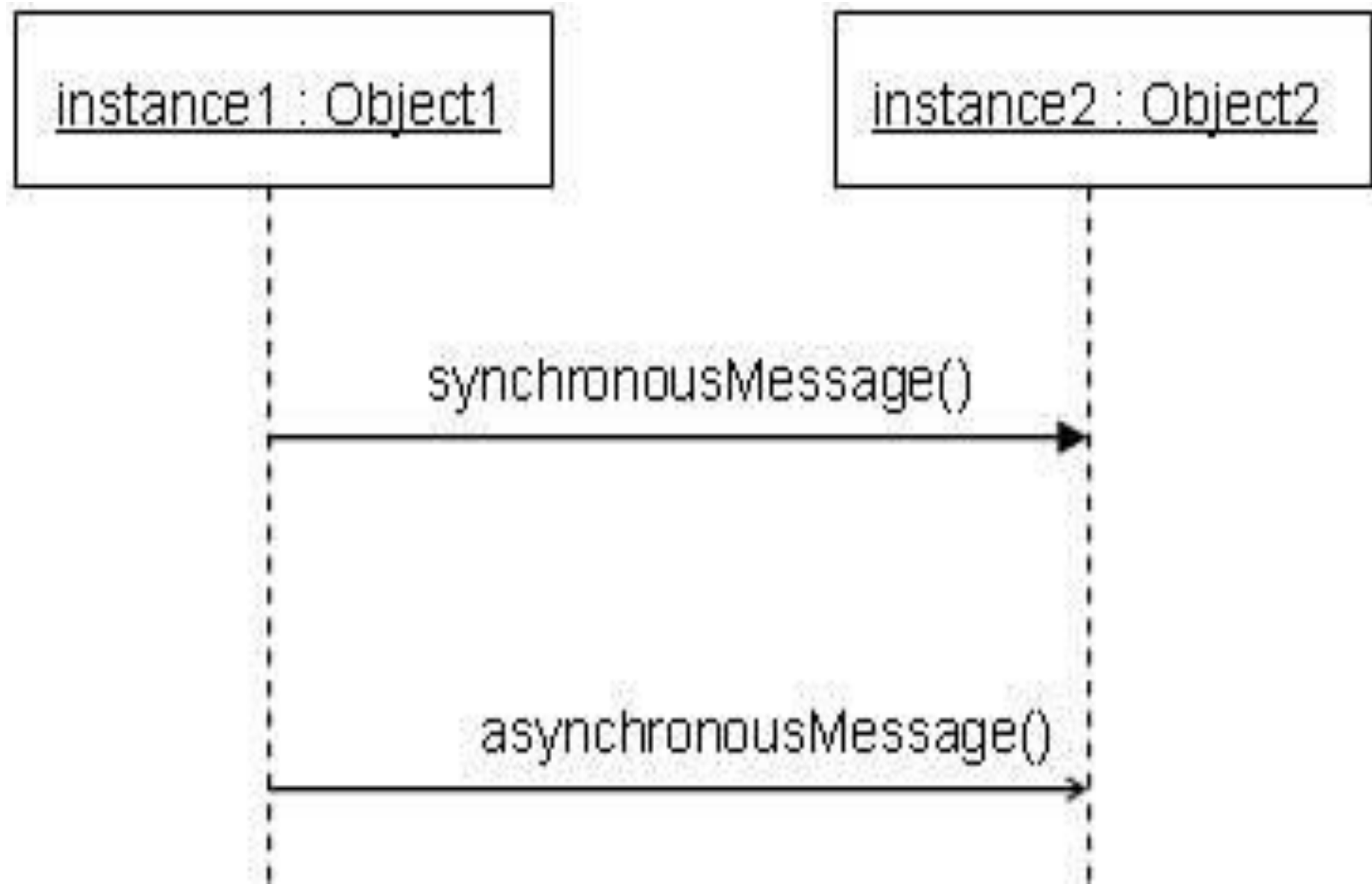
- **Synchronous messages** occur when the sender sends a request and **waits for a response** before continuing.
- **Asynchronous messages** allow the sender to send a request and **continue processing without waiting** for a reply.
- To draw an object calling itself, draw a message as normally, but instead of connecting it to another object, connect the message back to the object itself.
- The return messages are optional; a return message is drawn as a dotted line with an open arrowhead back to the originating lifeline, and above this dotted line the return value from the operation is placed.

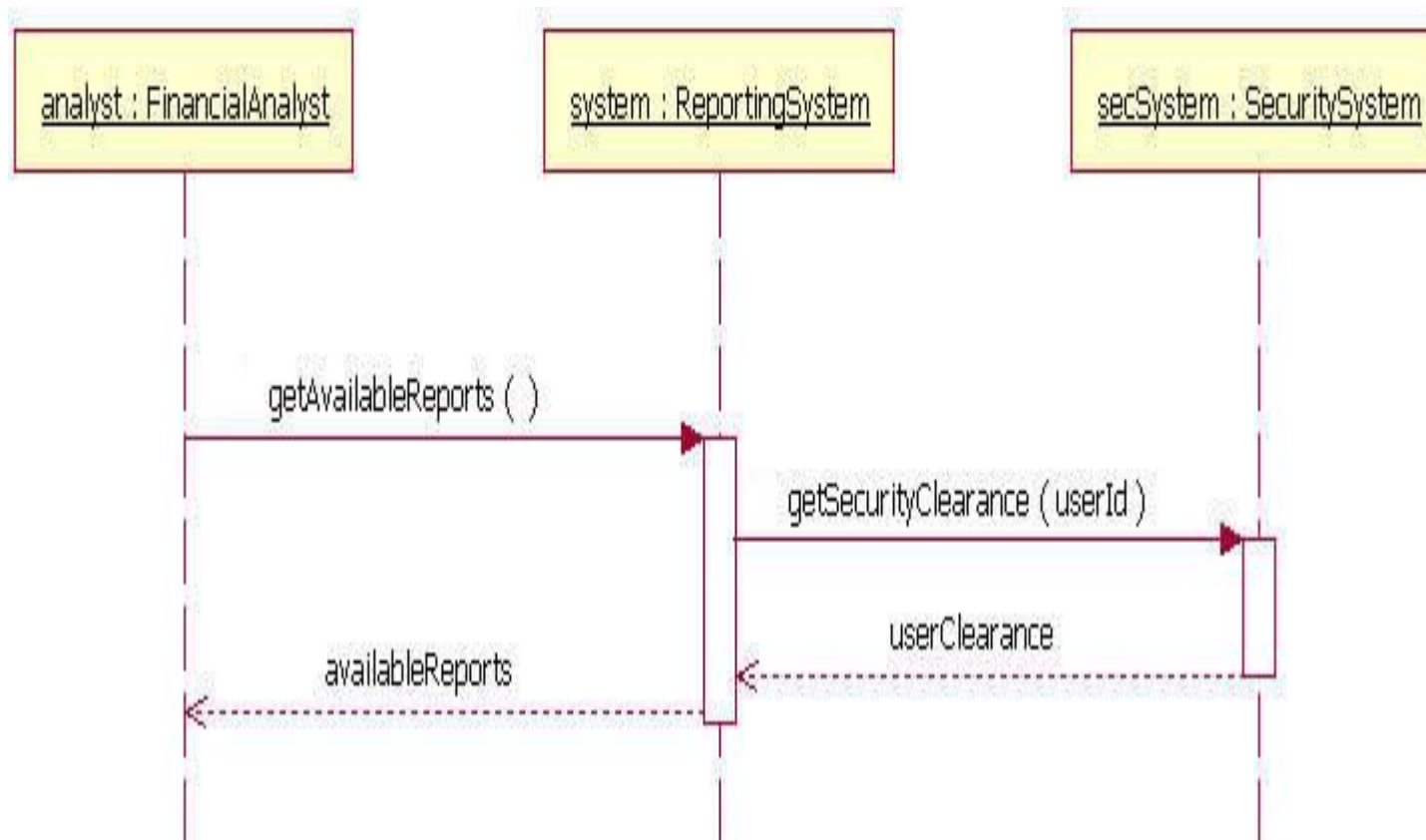
instance1 : Object1

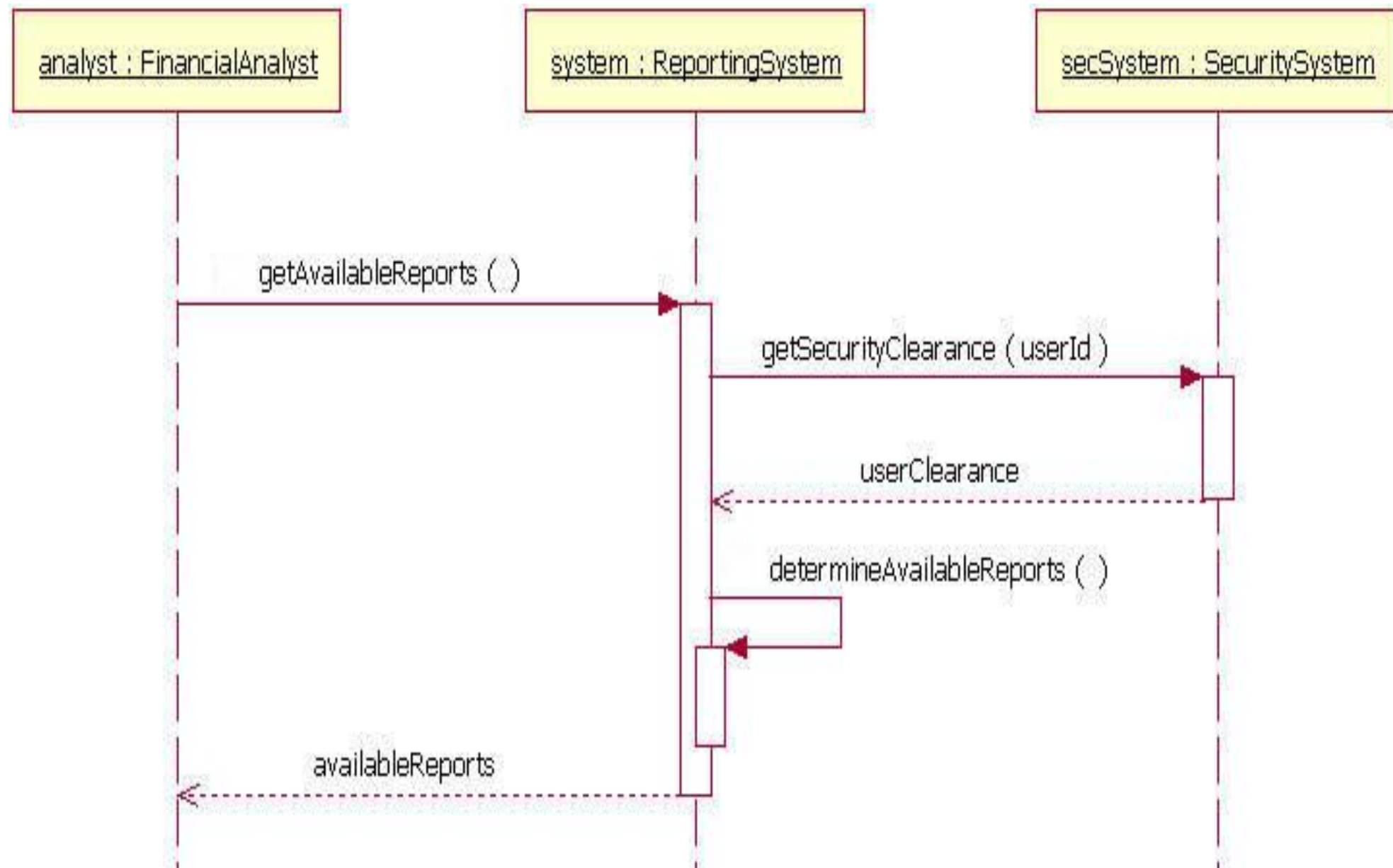
instance2 : Object2

synchronousMessage()

asynchronousMessage()

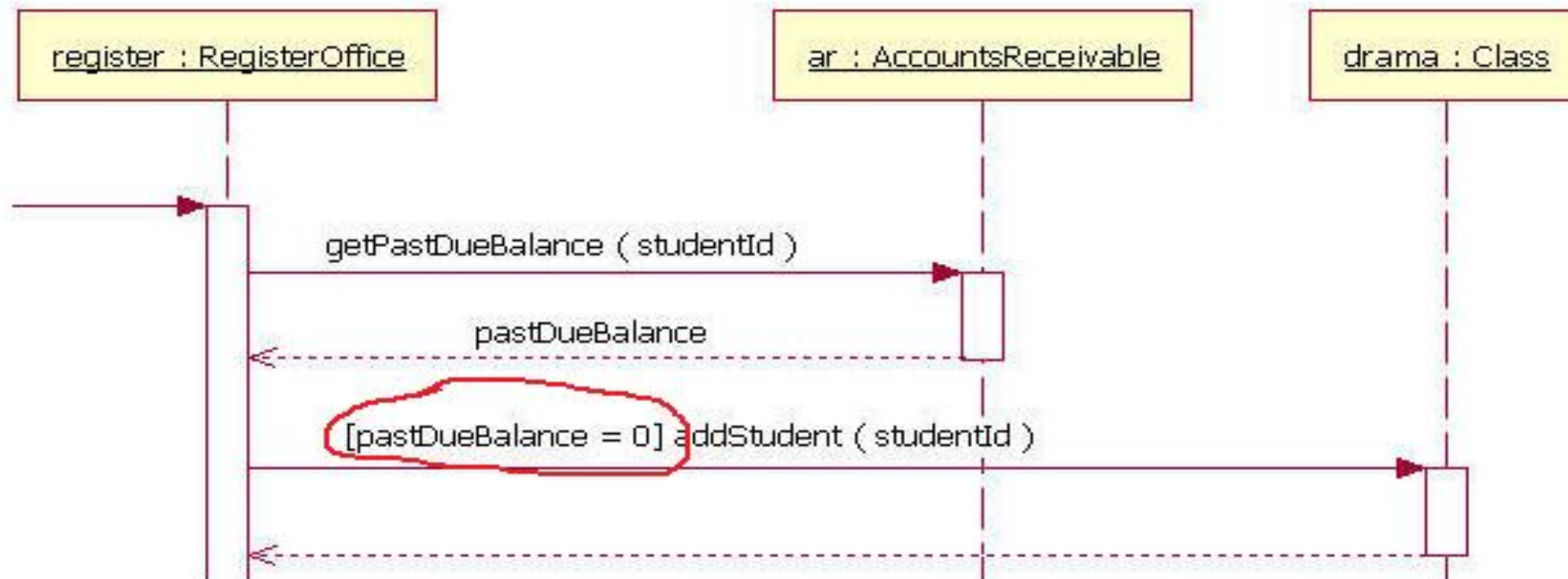






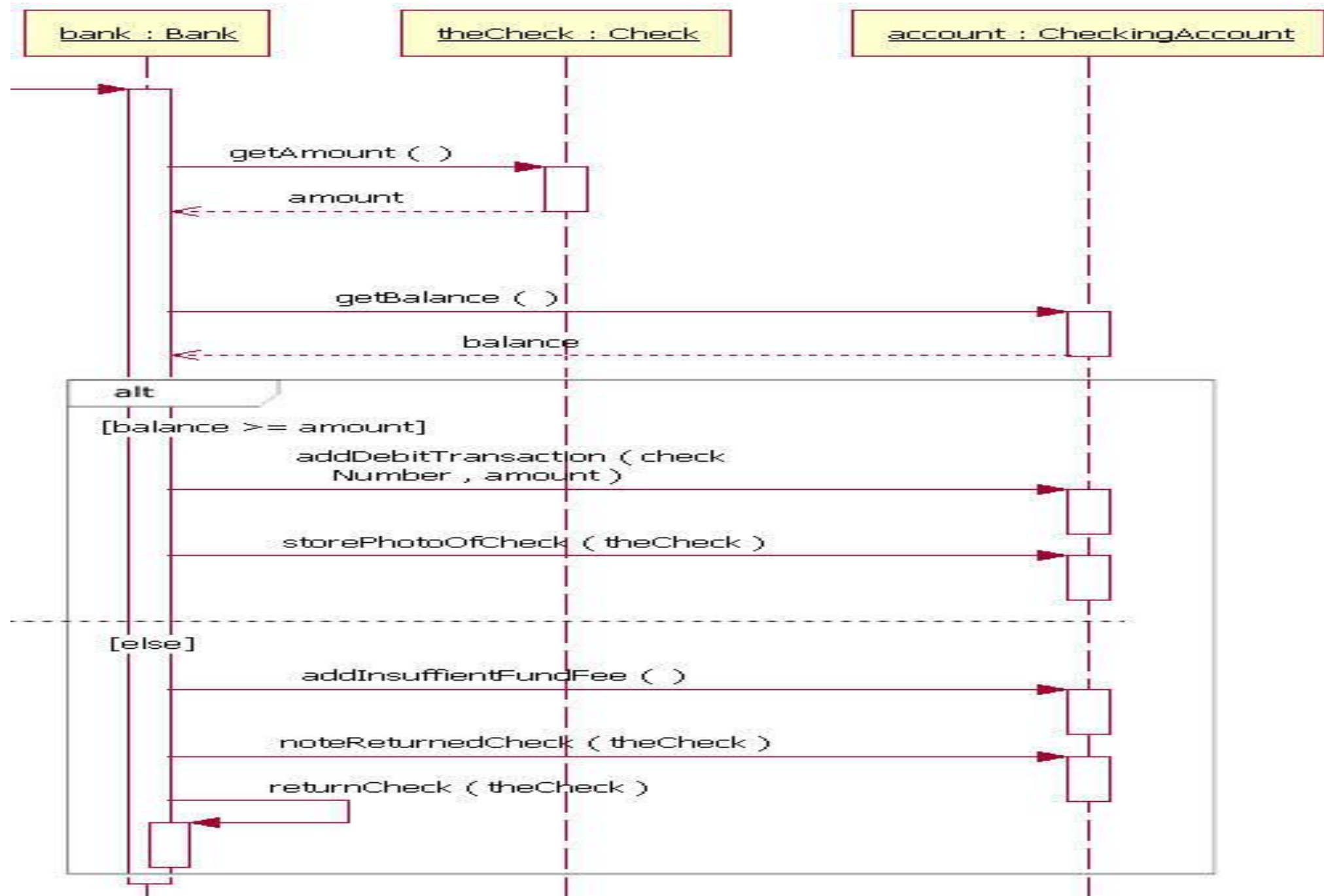
Guards

- When modelling object interactions, there will be times when a condition must be met for a message to be sent to the object. Guards are used throughout UML diagrams to control flow.



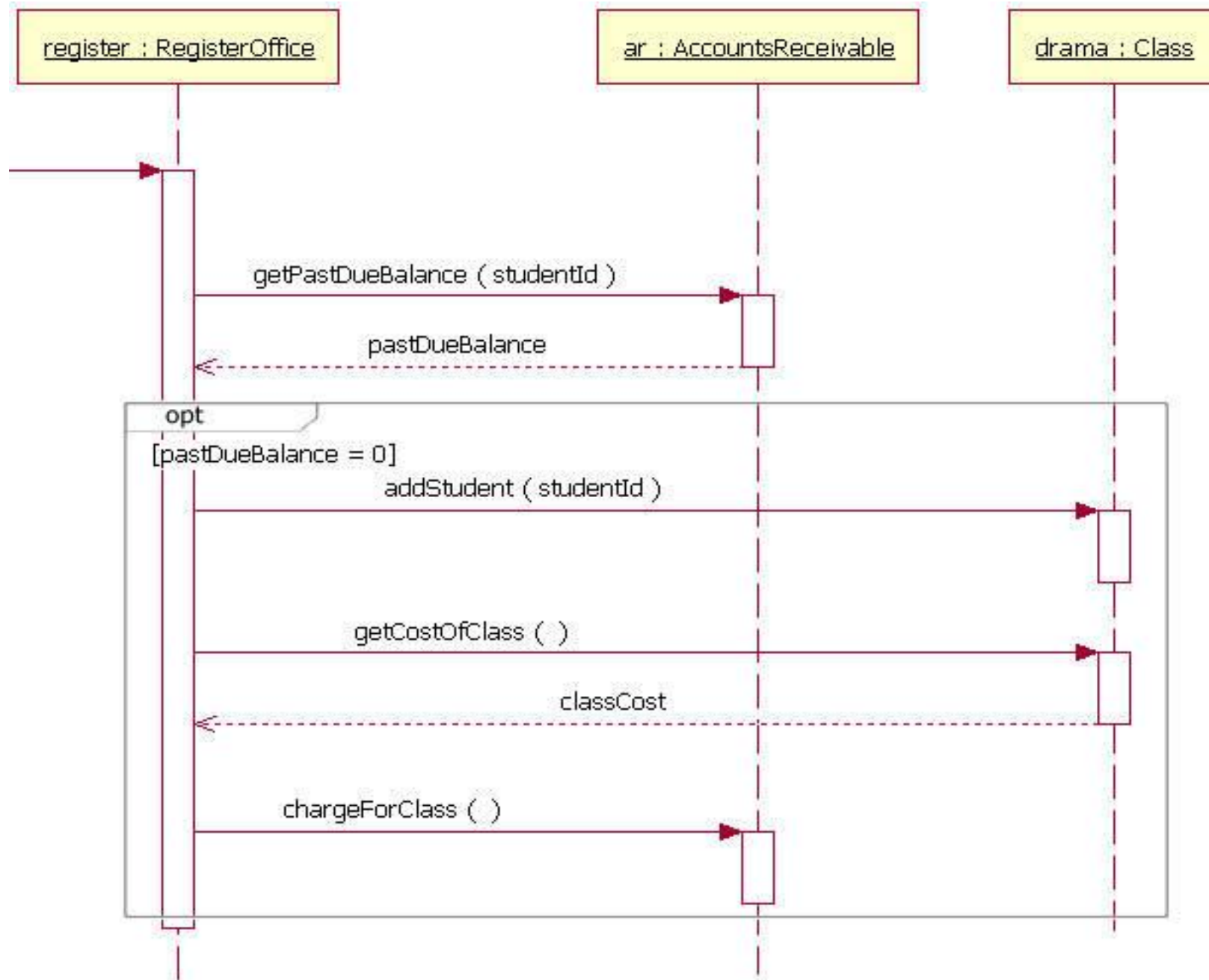
Alternatives

- Alternatives allow the modelling of the classic "if then else" logic (e.g., *if* I buy three items, *then* I get 20% off my purchase; *else* I get 10% off my purchase).
- An alternative combination fragment element is drawn using a frame. The word "alt" is placed inside the frame's name box.

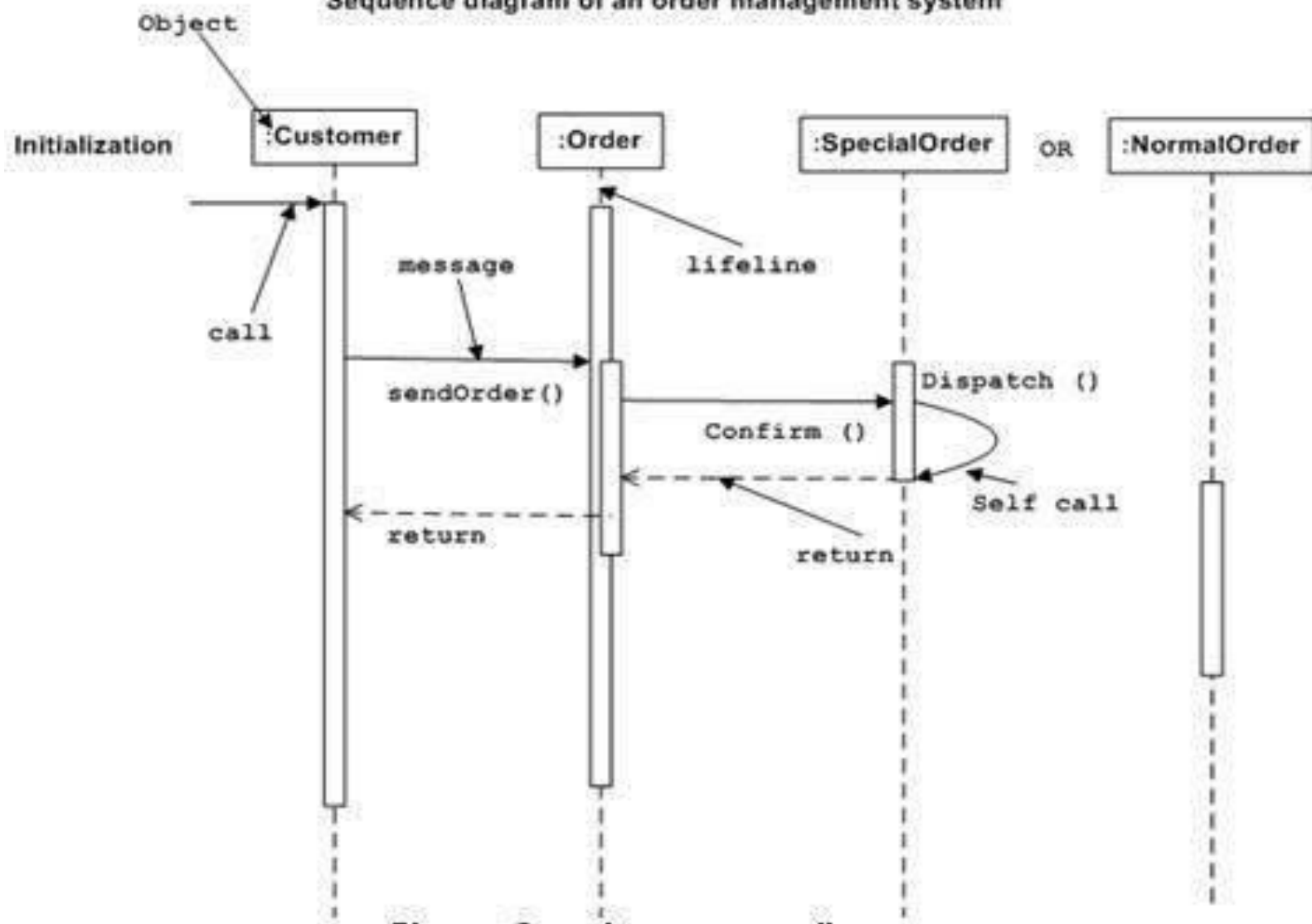


Option

- The option combination fragment is used to model a sequence that, given a certain condition, will occur; otherwise, the sequence does not occur. An option is used to model a simple "if then" statement
- To draw an option combination draw a frame. The text "opt" is placed inside the frame's name box, and in the frame's content area the option's guard is placed towards the top left corner on top of a lifeline. Then the option's sequence of messages is placed in the remainder of the frame's content area.



Sequence diagram of an order management system



Component Diagram

- Component diagrams are used to model the physical aspects of a system. Physical aspects are the elements such as executables, libraries, files, documents, etc. which reside in a node. It does not describe the functionality of the system but it describes the components used to make those functionalities.